

Mathematical Foundations of Machine Learning

Assignment 2 Solutions

Question 1 (18 points)

Consider the following four vectors in $\mathbb{R}^4$:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 5 \\ 0 \\ 4 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 4 \\ 3 \\ -3 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ -2 \end{bmatrix}$$

1. **Systematic Check (5 points)** Form a matrix $\mathbf{A}$ where these vectors are the columns. Perform Gaussian elimination to transform the matrix into row-echelon form.
2. **Pivot Criterion (4 points)** Identify the pivot columns of the matrix. Based on this, determine which vectors in the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ are linearly independent.
3. **Redundancy Identification (4 points)** Identify the redundant vector(s) in the set.
4. **Linear Combination (5 points)** Express the redundant vector you identified as an explicit linear combination of the linearly independent vectors.

Solution.

1. **Systematic Check: Gaussian Elimination** We begin by forming the matrix $\mathbf{A} \in \mathbb{R}^{4 \times 4}$ using the given vectors as columns:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 4 & 1 \\ -1 & 0 & 3 & 2 \\ 3 & 4 & -3 & -2 \end{bmatrix}$$

Applying elementary row operations:

(a) $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 + R_1$, and $R_4 \leftarrow R_4 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & -2 & -6 & -2 \end{bmatrix}$$

(b) $R_3 \leftarrow R_3 - 2R_2$ and $R_4 \leftarrow R_4 + 2R_2$:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{bmatrix}$$

(c) Swap R_3 and R_4 to achieve Row-Echelon Form:

$$\begin{bmatrix} \mathbf{1} & 2 & 1 & 0 \\ 0 & \mathbf{1} & 2 & 1 \\ 0 & 0 & \mathbf{-2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. The pivot elements are located in the **first, second, and third columns**. Following the pivot column criterion, the vectors corresponding to these columns are linearly independent:

Independent Set: $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$

3. Since the fourth column does not contain a pivot, the fourth vector is redundant:

Redundant Vector: $\mathbf{x}_4$

4. To find the coefficients α_i such that $\mathbf{x}_4 = \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \alpha_3\mathbf{x}_3$, we reduce the matrix to Reduced Row-Echelon Form:

$$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The final column reveals the coefficients:

$$\mathbf{x}_4 = -2\mathbf{x}_1 + 1\mathbf{x}_2 + 0\mathbf{x}_3$$

Verification:

$$-2 \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 5 \\ 0 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 4 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} -2 + 2 + 0 \\ -4 + 5 + 0 \\ 2 + 0 + 0 \\ -6 + 4 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ -2 \end{bmatrix} = \mathbf{x}_4$$

Question 2 (12 points)

Let $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4\}$ be a linearly independent set of vectors in $\mathbb{R}^n$. We define a new set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ as follows:

$$\begin{aligned}\mathbf{x}_1 &= \mathbf{a}_1 + 2\mathbf{a}_2 - \mathbf{a}_3 \\ \mathbf{x}_2 &= 2\mathbf{a}_1 + 5\mathbf{a}_2 + \mathbf{a}_4 \\ \mathbf{x}_3 &= \mathbf{a}_1 + 4\mathbf{a}_2 + 3\mathbf{a}_3 + 2\mathbf{a}_4 \\ \mathbf{x}_4 &= \mathbf{a}_2 + 2\mathbf{a}_3 + \mathbf{a}_4\end{aligned}$$

1. Represent this system in the form $\mathbf{x}_j = \sum_{i=1}^4 \alpha_{ij} \mathbf{a}_i$. Construct the coefficient matrix $\mathbf{C}$ such that $\mathbf{X} = \mathbf{AC}$, where the columns of $\mathbf{C}$ are the coefficient vectors. (3 points)
2. Without knowing the specific entries of the vectors $\mathbf{a}_i$, determine if the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ is linearly independent. You must justify your answer by reducing the coefficient matrix $\mathbf{C}$. (3 points)
3. Determine a basis for the space of all vectors $\mathbf{v} \in \mathbb{R}^4$ satisfying $\sum_{j=1}^4 v_j \mathbf{x}_j = \mathbf{0}$. (6 points)

Solution.

1. The coefficients α_{ij} form the columns of the coefficient matrix $\mathbf{C}$ such that $\mathbf{X} = \mathbf{AC}$:

$$\mathbf{C} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 4 & 1 \\ -1 & 0 & 3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

2. To determine if $\{\mathbf{x}_1, \dots, \mathbf{x}_4\}$ are linearly independent, we reduce $\mathbf{C}$ to Row-Echelon Form:

$$(a) \ R_2 \leftarrow R_2 - 2R_1 \text{ and } R_3 \leftarrow R_3 + R_1:$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$(b) \ R_3 \leftarrow R_3 - 2R_2 \text{ and } R_4 \leftarrow R_4 - R_2:$$

$$\text{REF} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Conclusion: The matrix $\mathbf{C}$ has a rank of 2 (only two pivot columns). Since $\text{rank}(\mathbf{C}) < 4$, the coefficient vectors are linearly dependent. Because the vectors $\mathbf{x}_j$ are independent if and only if their coefficient vectors are independent, the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ is **linearly dependent**.

3. To find the basis of the dependency space, we solve the homogeneous system $\mathbf{C}\mathbf{v} = \mathbf{0}$ using the Reduced Row-Echelon Form:

$$\begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot variables are v_1, v_2 and free variables are v_3, v_4 . We extract the equations:

$$\begin{aligned} v_1 - 3v_3 - 2v_4 &= 0 \implies v_1 = 3v_3 + 2v_4 \\ v_2 + 2v_3 + v_4 &= 0 \implies v_2 = -2v_3 - v_4 \end{aligned}$$

The general solution vector $\mathbf{v}$ is:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = v_3 \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + v_4 \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

Thus, a basis for all possible dependencies is:

$$B_{\text{dep}} = \left\{ \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Question 3 (20 points)

Consider a subspace $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\} \subset \mathbb{R}^4$, where:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{x}_5 = \begin{bmatrix} 3 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

1. Use the basis from pivot columns recipe to find a basis B for U . (8 points)
2. State the dimension $\dim(U)$. (4 points)

3. Let $\mathbf{v} = \begin{bmatrix} 4 \\ 3 \\ 7 \\ 1 \end{bmatrix}$. Verify that $\mathbf{v} \in U$ and express $\mathbf{v}$ as a linear combination of the basis vectors found in part (1). (8 points)

Solution.

1. We reduce the matrix $\mathbf{A}$ to Row-Echelon Form (REF):

(a) $R_3 \leftarrow R_3 - R_1$ and $R_4 \leftarrow R_4 - R_1$:

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & -1 & -1 & -1 & -2 \end{bmatrix}$$

(b) $R_3 \leftarrow R_3 - R_2$ and $R_4 \leftarrow R_4 + R_2$:

$$\text{REF} = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivots are in the **first and second columns**. The extracted basis is:

$$B = \{\mathbf{x}_1, \mathbf{x}_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} \right\}$$

2. The number of pivot columns is 2, therefore:

$$\dim(U) = 2$$

3. Solving $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = \mathbf{v}$:

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 7 \\ 1 \\ 1 \end{bmatrix}$$

From the second component, $c_2 = 3$. Substituting into the first component, $c_1 + 2(3) = 4 \implies c_1 = -2$. Thus:

$$\mathbf{v} = -2\mathbf{x}_1 + 3\mathbf{x}_2$$

Question 4 (20 points)

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 2 & 4 & -2 & 0 & 6 \\ 0 & 0 & 1 & -1 & 2 \\ 1 & 2 & 0 & 1 & 5 \end{bmatrix} \in \mathbb{R}^{4 \times 5}, \quad \mathbf{b}^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b}^{(2)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

1. (6 points) Use Gaussian elimination to reduce $\mathbf{A}$ to row-echelon form. Identify the pivot columns and compute $\text{rank}(\mathbf{A})$.
2. (4 points) Find a basis of the column space $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_5) \subseteq \mathbb{R}^4$ of $\mathbf{A}$ (where $\mathbf{a}_j$ denotes the j -th column of $\mathbf{A}$). State its dimension.
3. (4 points) Express each non-pivot column of $\mathbf{A}$ as a linear combination of the pivot columns.
4. (6 points) Use the rank condition to determine whether $\mathbf{A}\mathbf{x} = \mathbf{b}^{(1)}$ is solvable and whether $\mathbf{A}\mathbf{x} = \mathbf{b}^{(2)}$ is solvable. (You don't need to solve for $\mathbf{x}$.)

Solution:

1. Reduce $\mathbf{A}$:

$$\begin{aligned}
 & \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 2 & 4 & -2 & 0 & 6 \\ 0 & 0 & 1 & -1 & 2 \\ 1 & 2 & 0 & 1 & 5 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_4 \leftarrow R_4 - R_1} \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix} \\
 & \xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

There are 3 pivots, hence $\text{rank}(\mathbf{A}) = 3$. Pivot columns are 1, 3, 4.

2. A basis for the column space is given by the pivot columns of the *original* matrix:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_4 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Therefore, $\dim(\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_5)) = \text{rank}(\mathbf{A}) = 3$.

3. The non-pivot columns are $\mathbf{a}_2$ and $\mathbf{a}_5$. We can verify directly that

$$\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = 2\mathbf{a}_1,$$

and

$$\mathbf{a}_5 = \begin{bmatrix} 3 \\ 6 \\ 2 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = 5\mathbf{a}_1 + 2\mathbf{a}_3.$$

4. For $\mathbf{b}^{(1)}$, augment and eliminate similarly:

$$[\mathbf{A} \mathbf{b}^{(1)}] = \left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & 3 & 1 \\ 2 & 4 & -2 & 0 & 6 & 2 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 5 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_4 \leftarrow R_4 - R_1} \left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 2 & -1 \end{array} \right].$$

Continuing as before, no contradiction appears, and one obtains $\text{rank}([\mathbf{A} \mathbf{b}^{(1)}]) = 3 = \text{rank}(\mathbf{A})$, so $\mathbf{Ax} = \mathbf{b}^{(1)}$ is solvable.

For $\mathbf{b}^{(2)}$:

$$[\mathbf{A} \mathbf{b}^{(2)}] = \left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & 3 & 1 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 5 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 5 & 0 \end{array} \right].$$

The second row encodes $0 = -2$, so $\text{rank}([\mathbf{A} \mathbf{b}^{(2)}]) > \text{rank}(\mathbf{A})$ and the system is not solvable.

Question 5 (15 points)

Consider the ordered set $B_\lambda = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ in $\mathbb{R}^3$ where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ \lambda \end{bmatrix}.$$

- (5 points) For which values of λ is B_λ a basis of $\mathbb{R}^3$?
- (6 points) Let $\lambda = 3$ and $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Compute the coordinate vector $\hat{\mathbf{x}}$ of $\mathbf{x}$ with respect to B_3 .
- (4 points) For general $\lambda \neq 2$, let $\hat{\mathbf{z}} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$. Compute the vector $\mathbf{z} \in \mathbb{R}^3$ (in standard coordinates) whose coordinate vector with respect to B_λ equals $\hat{\mathbf{z}}$.

Solution:

- Form the matrix $\mathbf{M}_\lambda$ whose columns are $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$:

$$\mathbf{M}_\lambda = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & \lambda \end{bmatrix}.$$

By the pivot-column criterion, $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are linearly independent iff M_λ has a pivot in every column. Row-reduce:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & \lambda \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & \lambda - 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & \lambda - 2 \end{bmatrix}.$$

Therefore, there is a pivot in the third column iff $\lambda - 2 \neq 0$, i.e., iff $\lambda \neq 2$. Hence B_λ is a basis of $\mathbb{R}^3$ exactly when $\lambda \neq 2$.

2. For $\lambda = 3$, solve $\mathbf{x} = \beta_1 \mathbf{b}_1 + \beta_2 \mathbf{b}_2 + \beta_3 \mathbf{b}_3$:

$$\beta_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \beta_3 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Component-wise:

$$\beta_1 + \beta_3 = 1, \quad \beta_2 + \beta_3 = 2, \quad \beta_1 + \beta_2 + 3\beta_3 = 0.$$

From the first two equations, $\beta_1 = 1 - \beta_3$ and $\beta_2 = 2 - \beta_3$. Substitute into the third:

$$(1 - \beta_3) + (2 - \beta_3) + 3\beta_3 = 0 \implies 3 + \beta_3 = 0 \implies \beta_3 = -3.$$

Hence $\beta_1 = 4$ and $\beta_2 = 5$. Therefore,

$$\hat{\mathbf{x}} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -3 \end{bmatrix}.$$

3. If $\hat{\mathbf{z}} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, then by definition of coordinates in basis B_λ ,

$$\mathbf{z} = 1 \cdot \mathbf{b}_1 + (-2) \cdot \mathbf{b}_2 + 1 \cdot \mathbf{b}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \lambda \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ \lambda - 1 \end{bmatrix}.$$

Question 6 (15 points)

Let V be a 3-dimensional vector space with ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ and let W be a 4-dimensional vector space with ordered basis $C = (\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4)$, where the basis vectors of W (in standard coordinates of $\mathbb{R}^4$) are

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

A linear map $\Phi : V \rightarrow W$ is specified by (in C -coordinates):

$$\Phi(\mathbf{b}_1 + \mathbf{b}_2) = -\mathbf{c}_1 + 3\mathbf{c}_2 + 2\mathbf{c}_3 + 0\mathbf{c}_4, \quad \Phi(\mathbf{b}_2 - \mathbf{b}_3) = -2\mathbf{c}_1 + 4\mathbf{c}_2 - \mathbf{c}_3 - \mathbf{c}_4,$$

$$\Phi(2\mathbf{b}_1 - \mathbf{b}_2 + \mathbf{b}_3) = 4\mathbf{c}_1 - 4\mathbf{c}_2 + 5\mathbf{c}_3 - \mathbf{c}_4.$$

1. (6 points) Determine the coordinate vectors $[\Phi(\mathbf{b}_1)]_C$, $[\Phi(\mathbf{b}_2)]_C$, $[\Phi(\mathbf{b}_3)]_C$.
2. (3 points) Compute the transformation matrix $\mathbf{A}_\Phi$ of Φ with respect to the ordered bases B and C .
3. (3 points) Let $\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ be the coordinate vector of some $\mathbf{x} \in V$ with respect to B . Compute the coordinate vector $\hat{\mathbf{y}}$ of $\mathbf{y} = \Phi(\mathbf{x}) \in W$ with respect to C .
4. (3 points) Using your $\hat{\mathbf{y}}$ from part (3), compute $\mathbf{y}$ in standard coordinates of $\mathbb{R}^4$.

Solution:

1. Let

$$\mathbf{u}_1 = [\Phi(\mathbf{b}_1)]_C, \quad \mathbf{u}_2 = [\Phi(\mathbf{b}_2)]_C, \quad \mathbf{u}_3 = [\Phi(\mathbf{b}_3)]_C \in \mathbb{R}^4.$$

The given relations become

$$\mathbf{u}_1 + \mathbf{u}_2 = \begin{bmatrix} -1 \\ 3 \\ 2 \\ 0 \end{bmatrix} =: \mathbf{v}_A, \quad \mathbf{u}_2 - \mathbf{u}_3 = \begin{bmatrix} -2 \\ 4 \\ -1 \\ -1 \end{bmatrix} =: \mathbf{v}_B, \quad 2\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3 = \begin{bmatrix} 4 \\ -4 \\ 5 \\ -1 \end{bmatrix} =: \mathbf{v}_C.$$

From $\mathbf{u}_1 + \mathbf{u}_2 = \mathbf{v}_A$ we have $\mathbf{u}_2 = \mathbf{v}_A - \mathbf{u}_1$. From $\mathbf{u}_2 - \mathbf{u}_3 = \mathbf{v}_B$ we have $\mathbf{u}_3 = \mathbf{u}_2 - \mathbf{v}_B = \mathbf{v}_A - \mathbf{u}_1 - \mathbf{v}_B$. Substitute into $2\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3 = \mathbf{v}_C$:

$$2\mathbf{u}_1 - (\mathbf{v}_A - \mathbf{u}_1) + (\mathbf{v}_A - \mathbf{u}_1 - \mathbf{v}_B) = \mathbf{v}_C \implies 2\mathbf{u}_1 - \mathbf{v}_B = \mathbf{v}_C \implies 2\mathbf{u}_1 = \mathbf{v}_B + \mathbf{v}_C.$$

Hence

$$\mathbf{u}_1 = \frac{1}{2} \left(\begin{bmatrix} -2 \\ 4 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 4 \\ -4 \\ 5 \\ -1 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 2 \\ 0 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}.$$

Then

$$\mathbf{u}_2 = \mathbf{v}_A - \mathbf{u}_1 = \begin{bmatrix} -1 \\ 3 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \mathbf{u}_2 - \mathbf{v}_B = \begin{bmatrix} -2 \\ 3 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -2 \\ 4 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}.$$

Therefore,

$$[\Phi(\mathbf{b}_1)]_C = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}, \quad [\Phi(\mathbf{b}_2)]_C = \begin{bmatrix} -2 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \quad [\Phi(\mathbf{b}_3)]_C = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}.$$

2. The transformation matrix $\mathbf{A}_\Phi$ has columns $[\Phi(\mathbf{b}_j)]_C$:

$$\mathbf{A}_\Phi = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 3 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix}.$$

3. Using $\hat{\mathbf{y}} = \mathbf{A}_\Phi \hat{\mathbf{x}}$ with $\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$:

$$\hat{\mathbf{y}} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 3 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 + 2 + 0 \\ 0 - 3 - 2 \\ 6 + 0 + 2 \\ -3 - 1 + 4 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 8 \\ 0 \end{bmatrix}.$$

4. Since $\hat{\mathbf{y}} = [\mathbf{y}]_C$, we have

$$\mathbf{y} = 5\mathbf{c}_1 - 5\mathbf{c}_2 + 8\mathbf{c}_3 + 0\mathbf{c}_4.$$

Substitute the given $\mathbf{c}_i$ (standard coordinates in $\mathbb{R}^4$):

$$\mathbf{y} = 5 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - 5 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 8 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ 8 \\ 0 \end{bmatrix}.$$

Bonus Question (10 points)

Consider the linear transformation on $\mathbb{R}^2$ given by $\mathbf{y} = \mathbf{A}\mathbf{x}$ with

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

- (3 points) Compute $\mathbf{A}\mathbf{e}_1$, $\mathbf{A}\mathbf{e}_2$, and $\mathbf{A} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
- (3 points) Note that the vectors in part 1, along with the vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ correspond to the 4 vertices of a unit square. Describe how the linear transformation changes the unit square by words or by drawing the figure.

3. (4 points) Briefly describe (in words) the general geometric effect of $\mathbf{A}$ (e.g., which directions are stretched and how).

Solution:

1. Compute:

$$\mathbf{A}\mathbf{e}_1 = \mathbf{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{A}\mathbf{e}_2 = \mathbf{A} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{A} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

2. The unit square becomes a parallelogram with one edge fixed along the x -axis and the vertical edges tilted to the right.
3. Since $y_1 = x_1 + x_2$ and $y_2 = 2x_2$, the transformation doubles the vertical component (stretching in the $\mathbf{e}_2$ direction by a factor of 2) and shears horizontally by adding the original x_2 into the x_1 -coordinate.